

MUHAMMED FARIS MUKTHAR M V

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SUMMARY

AI Developer with a Master's in Computer Science (Data Analytics) specialising in building complete, production-ready AI-powered systems — from model training and LLM/RAG integration to full-stack web applications, Android apps, and cloud deployment. Experienced in GenAI (RAG, LangChain, LLMs), computer vision (YOLOv8), and NLP with a strong ability to own the entire product lifecycle independently. Skilled in Next.js, Supabase, AWS, and Docker, with hands-on experience shipping AI products that serve real users across web and mobile platforms.

TECHNICAL SKILLS

AI / GenAI: LLMs, LangChain, RAG, YOLOv8, LayoutLM, RoBERTa, Siamese Networks, CNN/RNN/LSTM
Web & Mobile: Next.js, Node.js, REST APIs, Supabase (PostgreSQL), Android Studio
Cloud & DevOps: AWS (EC2, S3), Docker, Microsoft Azure, Model Deployment
Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy
Data & Analytics: SQL, Predictive Modelling, EDA, Tableau, Power BI
Languages: Python, JavaScript, SQL, Embedded C/C++, MicroPython, HTML/CSS
Edge Hardware: Raspberry Pi, Jetson Nano, ESP32, Arduino

EXPERIENCE

R&D AI Developer – Innovations

Oct 2025 – Present

Digital University Kerala

Technopark, Trivandrum

- Built a full-stack physical education management system combining IoT hardware, computer vision, and AI analytics for Kerala state schools.
- Independently designed and delivered a **children's health monitoring AI platform** — an end-to-end product spanning a full-stack web app, Android mobile app, IoT hardware integration, computer vision inference, and a cloud analytics dashboard.
- Built a **hierarchical multi-user web application** (Next.js, Supabase/PostgreSQL, AWS EC2/S3) with role-based access for super admins, teachers, and students; super admins assign activity drills, teachers monitor and give live feedback via mobile.
- Developed an **Android mobile application** (Android Studio, AI-assisted development) connecting to IoT hardware over Bluetooth/Wi-Fi, enabling teachers to control drills and upload real-time student feedback to the cloud database.
- Integrated **YOLOv8 computer vision** on Raspberry Pi for automated student activity detection and performance tracking, with results stored in Supabase for analytics and reporting to stakeholders.
- Building a **RAG-based student personal health chatbot** using LangChain and LLMs, grounded in real student health and performance data to deliver personalised AI-driven recommendations.
- Managed the **complete product lifecycle** independently: system architecture, backend APIs (Node.js), database schema design, cloud hosting (AWS), mobile app, and ML model deployment.
- Containerised ML services and backend using **Docker** for scalable, reproducible production deployments.

R&D Intern – AI & Full-Stack Systems

Apr 2025 – Oct 2025

Cisco thinQbator-DUK

Technopark, Trivandrum

- Integrated ML models with Raspberry Pi for real-time edge AI detection and classification tasks.
- Built a responsive Next.js dashboard streaming live IoT sensor data and ML model predictions in real time.
- Developed IoT data pipelines connecting Raspberry Pi, Arduino, and ESP32 hardware to cloud backend services.

Consultant – Machine Learning Engineering

Sep 2024 – Dec 2024

Cleareye.ai

Technopark Phase I, Trivandrum

- Fine-tuned **RoBERTa** for Trade Finance NLP tasks and engineered **LayoutLM + OpenCV** document analysis pipelines, improving end-to-end document automation accuracy.
- Built **Siamese network** signature verification and **YOLOv8** seal detection systems integrated into production document processing workflows.
- Created synthetic datasets to overcome data scarcity and improve model performance across document types.

Machine Learning Intern

Cleareye.ai

May 2024 – Aug 2024

Technopark Phase I, Trivandrum

- Fine-tuned LayoutLM for complex document extraction and built content/metadata classification pipelines.
- Generated synthetic training data to expand datasets and improve model generalisation.

Machine Learning Research Intern

Digital University Kerala

Feb 2024 – May 2024

Technopark, Trivandrum

- Automated document sanctity checks using structural OCR, LayoutLM layout analysis, and YOLO-based detection pipelines.

PROJECTS

Document Extraction Chatbot | *LLMs, RAG, LangChain, TensorFlow*

- Government order chatbot with end-to-end RAG pipeline — document ingestion, vector retrieval, and context-aware LLM response generation.

Plant Disease Detection App | *YOLOv8, OpenCV, Python*

- Real-time plant disease detection using YOLOv8, surfacing actionable insights for farmers and agronomists.

Customer Retail Forecasting | *ML, Time Series, Scikit-learn*

- Purchasing trend analysis and predictive modelling delivering actionable business intelligence for retail decision-making.

EDUCATION

Master of Computer Science in Data Analytics

Digital University Kerala

Aug 2022 – Jul 2024

Trivandrum, Kerala

Bachelor of Science in Mathematics

Farook College, Calicut University

Jun 2019 – Mar 2022

Calicut, Kerala

ACTIVITIES

UGC-NET — Qualified for admission to Ph.D. ONLY

2026

NCC 'C' Certificate Holder

2022

National Cadet Corps (NCC), India — Company Quartermaster Sergeant

2020 – 2022